Zirui Chen, Ph.D. Candidate

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Education

2024 – Present — **Ph.D. Candidate in Computer Science and Technology**, Westlake University
Joint program with Zhejiang University; WINDY Lab, School of Engineering
Supervisor: Prof. Shiyu Zhao
Tentative thesis: *Collective Behavior Clone via Neural Network*

2021 – 2024 — **M.Eng. in Control Theory and Engineering**, Beihang University
The Seventh Division, School of Automation Science and Electrical Engineering
Supervisor: Prof. Zongyu Zuo
Thesis: *Nonsingular Guiding Vector Field for Geometric Path Following Control*

2017 – 2021 — **B.Eng. in Automation**, Beihang University
School of Automation Science and Electrical Engineering
Thesis: *Path Following Control for Airship*

Research Publications

1. Y. Feng, Z. Zhang, Z. Liang, N. Zhang, F. Zhang, S. Cao, S. Zhang, and **Z. Chen**, "Completing the response space: Unifying rejection and multi-interval video grounding via cardinality-rectified grpo," in *2026 IEEE International Conference on Multimedia and Expo (ICME)*, Best Paper Award; corresponding author, Bangkok, Thailand, Jul. 2026.
2. **Z. Chen**, S. Guo, and S. Zhao, "Guiding vector field generation via score-based diffusion model," in *2026 IEEE International Conference on Robotics and Automation (ICRA)*, Vienna, Austria, Jun. 2026.
3. **Z. Chen**, L. Zhang, G. Chen, et al., "Cyborg-swarm cooperation and game via affective-based brain–machine interface," *National Science Review*, vol. 13, no. 13, nwag313, May 2026. DOI: 10.1093/nsr/nwag313.
4. **Z. Chen**, Z. Zuo, G. Wang, P. Li, and Y. Liu, "Discrete data-based frequency-domain closed planar path following method," *IEEE/ASME Transactions on Mechatronics*, vol. 30, no. 6, pp. 5330–5341, Dec. 2025. DOI: 10.1109/tmech.2025.3566540.
5. **Z. Chen** and Z. Zuo, "Non-singular cooperative guiding vector field under a homotopy equivalence transformation," *Automatica*, vol. 171, p. 111 962, Jan. 2025. DOI: 10.1016/j.automatica.2024.111962.
6. **Z. Chen**, J. Tang, and Z. Zuo, "A novel prescribed-performance path-following problem for non-holonomic vehicles," *IEEE/CAA Journal of Automatica Sinica*, vol. 11, no. 6, pp. 1476–1484, Jun. 2024. DOI: 10.1109/jas.2024.124311.

Research Interests & Skills

Research — Robot learning, generative models, guiding vector fields, differential geometry for deep learning, nonlinear control, and reinforcement learning.

Languages — Mandarin Chinese (native); English (fluent, IELTS 6.5).

Academic Activities

Selected Presentations

2026 — **ICME 2026**, Bangkok, Thailand — Oral presentation of *Completing the Response Space: Unifying Rejection and Multi-Interval Video Grounding via Cardinality-Rectified GRPO*.

- **ICRA 2026**, Vienna, Austria — Poster presentation of *Guiding Vector Field Generation via Score-based Diffusion Model*.

2025 — **IROS 2025 Workshop** — Invited talk, *Exploring Vector Field Construction Methods for Different Types of Paths*.

Teaching & Outreach

2025 – Present — Science Communicator, Westlake University Science Outreach Team; developed and delivered *Inspirations from Geocentrism*.

2025 — Teaching Assistant, *Mathematical Foundations of Reinforcement Learning*, Westlake University.

2022 – 2023 — Teaching Assistant, *Principles of Automatic Control*, Beihang University.

2021 — Teaching Assistant, *Theoretical Mechanics*, Beihang University.

2020 — Teaching Assistant, *College English: Critical Thinking and Academic Writing*, Beihang University.

Academic Service

2024 – Present — Reviewer for *The International Journal of Robotics Research*, *IEEE Robotics and Automation Letters*, *Automatica*, *IEEE Transactions on Automatic Control*, *IEEE Transactions on Industrial Informatics*, and *Journal of the Astronautical Sciences*.

Awards & Honors

2026 — **ICME 2026 Best Paper Award**, Bangkok, Thailand.

2025 — **CAA Outstanding Master's Thesis Award**, Beijing, China.

2021 — **Best Teaching Assistant Award**, Beihang College, Beihang University.